

1. Administrative & Study Identification

Full Study Title: A Phase 3, Open-label Extension Study to Evaluate the Safety and Efficacy of MK-0616 in Adults With Hypercholesterolemia **Short Title/Acronym:** CORALreef Extension (MK-0616-019) **Protocol Number:** MK-0616-019 **NCT Number:** NCT06492291 **Sponsor Name:** Merck Sharp & Dohme LLC **Investigational Product:** Enlicitide Decanoate (MK-0616) / Oral PCSK9 Inhibitor **Phase of Development:** Phase 3 **Medical Monitor:** Merck Sharp & Dohme LLC Clinical Operations Lead

2. Study Synopsis & Rationale

2.1 Study Objective Primary Objective: To evaluate the longer-term safety, tolerability, and efficacy of enlicitide decanoate (MK-0616) in adults with hypercholesterolemia who completed a parent trial. **Secondary Objectives:** To evaluate the percentage of participants achieving target LDL-C goals (e.g., <70 mg/dL and <55 mg/dL) with a $\geq 50\%$ reduction from baseline, and to evaluate mean percent changes in non-HDL-C and ApoB.

2.2 Background & Rationale To address the ongoing risk of atherosclerotic cardiovascular disease (ASCVD) resulting from elevated low-density lipoprotein cholesterol (LDL-C). While injectable PCSK9 inhibitors are highly effective, they are underutilized in clinical practice. Enlicitide decanoate is designed to provide the same biological mechanism of action in a daily oral pill. This CORALreef Extension study evaluates the long-term safety and effectiveness of the intervention for participants who successfully completed one of the parent phase 3 trials (MK-0616-013, MK-0616-017, or MK-0616-018).

3. Study Design & Methodology

3.1 Design Framework Study Type: Interventional **Control Type:** Single-arm (Open-label extension) **Blinding:** Open-label (Both participants and researchers know the treatment being administered) **Randomization:** N/A

3.2 Planned Enrollment & Duration Target $\$N\$:$ Approximately 3,000 participants **Number of Sites:** Global / Multicenter **Estimated Study Start:** 05-Aug-2024 **Estimated Primary Completion:** 13-Oct-2028 **Estimated Study Completion:** 13-Oct-2028

4. Subject Selection (Eligibility Criteria)

Demographics & Population: - Age: ≥ 18 years - Sex: All - Healthy Volunteers: No

4.1 Inclusion Criteria 1. Has hypercholesterolemia. 2. Completed an elicited decanoate parent study (MK-0616-013, MK-0616-017, or MK-0616-018) per protocol, including final assessments. 3. Had an overall study intervention compliance ≥ 80% while participating in their parent study.

4.2 Exclusion Criteria 1. Has discontinued study intervention in their parent study. 2. Has an ongoing serious adverse event considered related to study intervention by the investigator in the parent study. 3. Is currently participating in an interventional clinical study other than an elicited decanoate parent study, or is planning to begin treatment with a nonstudy PCSK9 inhibitor.

5. Intervention & Treatment Plan

5.1 Investigational Regimen Dosage/Frequency: Elicited Decanoate once a day. Route of Administration: Oral (tablet). Must be taken on an empty stomach; participants must not eat or drink anything (except small amounts of water) for 8 hours before and 30 minutes after taking the medicine. Duration of Treatment: Up to 3 years depending on the time of entry into the extension trial.

5.2 Concomitant & Prohibited Therapy Permitted: Stable background lipid-lowering therapies (e.g., statins) continued from the parent trial. Prohibited: Use of nonstudy PCSK9 inhibitors or concurrent enrollment in another interventional clinical trial.

6. Study Timeline & Visit Schedule

Visit ID	Window (Days)	Key Activities/Procedures
1	Screening/Baseline	Parent Roll-over
2	Enrollment confirmation, verification of ≥ 80% parent study compliance, baseline safety evaluation.	Interim
3	Ongoing	Routine blood and urine tests, physical examinations, AE monitoring, LDL-C tracking.
4	End of Study	Up to Year 3
5	Final physical examination, safety labs, efficacy assessment, and IP accountability.	

7. Outcome Measures (Endpoints)

7.1 Primary Endpoint Definition: Long-term safety and tolerability assessed by the number of participants experiencing one or more Adverse Events (AEs) or discontinuing study drug due

to AEs; Long-term efficacy measured by mean percent change in LDL-C from baseline. **Measurement Method:** Blood lipid panel assessments and standardized clinical safety evaluations.

7.2 Secondary & Exploratory Endpoints * Percentage of participants with LDL-C <70 mg/dL and $\geq 50\%$ reduction from parent study baseline. * Percentage of participants with LDL-C <55 mg/dL and $\geq 50\%$ reduction from parent study baseline. * Mean percent changes in non-HDL-C, Apolipoprotein B (ApoB), and Lipoprotein(a) [Lp(a)].

8. Safety & Adverse Event Reporting

Adverse Event (AE) Monitoring: Standard collection via physical examinations, vital signs, and laboratory (blood/urine) tests conducted throughout the extension trial. **Serious Adverse Events (SAEs):** Investigated comprehensively, especially considering any ongoing drug-related SAEs from the parent trials. Reporting timeline typically 24 hours. **Data Monitoring Committee (DMC):** Overseen by the Sponsor's safety review committees for the duration of the multi-year open-label extension.

9. Statistical Analysis Plan (SAP) Summary

Analysis Populations: Safety Set (all participants receiving ≥ 1 dose in the extension phase); Intent-to-Treat (ITT) using baseline data from the parent study for long-term efficacy analysis. **Sample Size Justification:** ~3,000 participants. Powered based on the projected roll-over rates from the collective parent CORALreef trials (MK-0616-013, 017, and 018). **Handling of Missing Data:** Standard methodologies (e.g., Multiple Imputation) adapted for long-term, open-label, repeated-measure data.

10. Quality Assurance & Regulatory Compliance

GCP Compliance: This study will be conducted in accordance with Good Clinical Practice (GCP) and the Declaration of Helsinki. **Monitoring Plan:** Scheduled onsite monitoring and central data review to ensure strict compliance with the dosing administration protocol (fasting windows). **Audit Trail:** Standard electronic case report form (eCRF) locking and query resolution processes.

11. Study Identifiers & Governance

Primary ID: MK-0616-019 **Registry Number:** NCT06492291 **Sponsor/Lead Organization:** Merck Sharp & Dohme LLC **Study Title:** CORALreef Extension **Official Regulatory Title:** A Phase 3, Open-label

12. Clinical Framework (Hypercholesterolemia Specific)

Primary Condition: Hypercholesterolemia **Diagnosis Threshold:** Established elevated LDL-C levels qualifying participants for parent Phase 3 trials. **Medical Methodology:** Daily oral PCSK9 inhibition (Enlicitide Decanoate). **Optimization Target:** Sustained LDL-C reduction targeting levels <70 mg/dL or <55 mg/dL depending on ASCVD risk.

13. Study Procedures & Methodology

Management Pathway: Long-term open-label follow-up extending from blinded parent trials. **Education Protocol (HCP):** Site training on verifying patient adherence to complex fasting requirements surrounding drug administration. **Education Protocol (Patient):** Strict guidance on the 8-hour pre-dose and 30-minute post-dose fasting requirements (only small amounts of water permitted). **Quality Audit Mechanism:** Medication accountability and compliance tracking (≥ 80% adherence required).

14. Observation & Follow-Up Schedule

Total Duration: Up to 3 years (36 months) **Onsite Visit Cadence:** Regular scheduled intervals bridging the 3-year extension period. **Remote Monitoring Frequency:** Supplemental remote contacts utilized per standard protocol for AE reporting. **Checklist Review Interval:** Regular reconciliation of dosing diaries to ensure fasting compliance.

15. Sign-off & Approval

Role	Name	Signature	Date	:--	:--	:--	:--
Principal Investigator	[Name]	_____	[DD-MMM-YYYY]				
Clinical Operations Lead	[Name]	_____	[DD-MMM-YYYY]		Lead		
Statistician	[Name]	_____	[DD-MMM-YYYY]				